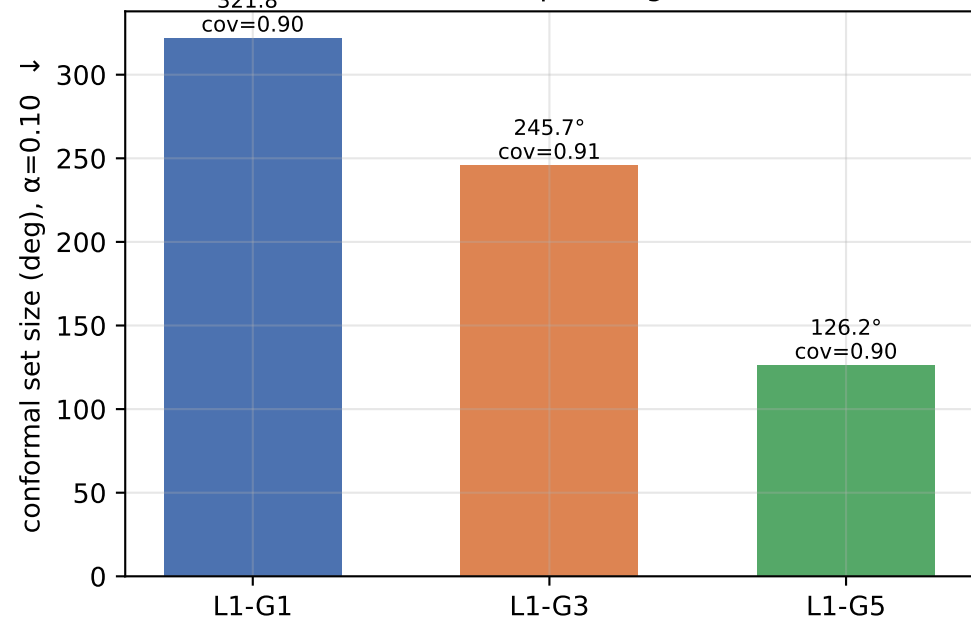


Fig.5 Confidence consistency: conformal set size vs geometry + retained negatives (model-known simulated)

(a) OCS-only conformal set size tightens with geometry
(current simulated split, target cov=0.90)



(b) negative-observation checklist

Negative observations (must retain):

- image_only conformal coverage $\approx 0.83-0.89$ (under target 0.90)
- neural margin risk-coverage nearly flat (weak confidence discrimination)
- P-EXT yaw-block still collapses ($\approx 146-157^\circ$)
- joint gain ceiling-limited; final-checkpoint sensitivity (G5 joint moderate hit@30=0.189)
- oracle is upper bound, not unsupervised selection of the correct branch